Regime-Adaptive Multimodal Transformer for Cross-Market Equity Forecasting

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Overview

Agenda

Problem Statement

TRADITIONAL MODEL FAILS TO UNDERSTAND MARKET COMPLEXITY

Financial markets generate massive volumes of heterogeneous data, from tick-level prices to unstructured news feeds, yet conventional forecasting methods rely on rigid statistical assumptions that break down under volatility. This gap between data richness and model expressiveness leads to poor risk estimates, missed signals, and costly allocation errors.

Data heterogeneity

Financialdatasets spanstructured time series, order-book microstructure, and unstructured text. Classical models cannot fuse these modalities into a unified predictive framework.

Non-stationarity

Market regimes shift abruptly. Models trained on historical distributions degrade when volatility spikes or correlations break, producing unreliable forecasts precisely when accuracy matters most.

Interpretability gap

Deep learning approached improve accuracy but operate as black boxes. Regulatory and fiduciary requirements demand explainable outputs, limiting real- world adoption in portfolio management.

Literature Review

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- Cross-sectional momentum

  Jegadeesh & Titman (1993) established that past 3-12 month winners continue winning; Asness et al. (2013) confirmed the factor persists across markets

- Regime detection

  Hamilton (1989) introduced HMMs for financial time series; Guidolin & Timmermann (2007) applied regime-switching to portfolio allocation

- Transformers for time series

  Vaswani et al. (2017) introduced the architecture; Lim et al. (2021) adapted it for multi-horizon forecasting with the Temporal Fusion Transformer

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Research journey

Six iterations — two failures, one pivot, one working strategy.

01

XGBoost, daily returns

Predicted next-day return on NIFTY 200. Signal-to-noise too low at daily horizon. Abandoned.

02

LSTM, daily returns

Sequence model on the same daily task. Underperformed XGBoost. Confirmed the target was the problem, not the model.

03

Pivot to monthly alpha

Retargeted to 21-day forward alpha vs NIFTY. XGBoost and LSTM retrained. Signal emerged but ranking quality was weak.

04

RAMT transformer

Multimodal encoder, regime cross-attention, MoE, tournament ranking loss. Prediction collapse on test set. Validation correlation went negative.

05

LightGBM diagnostic

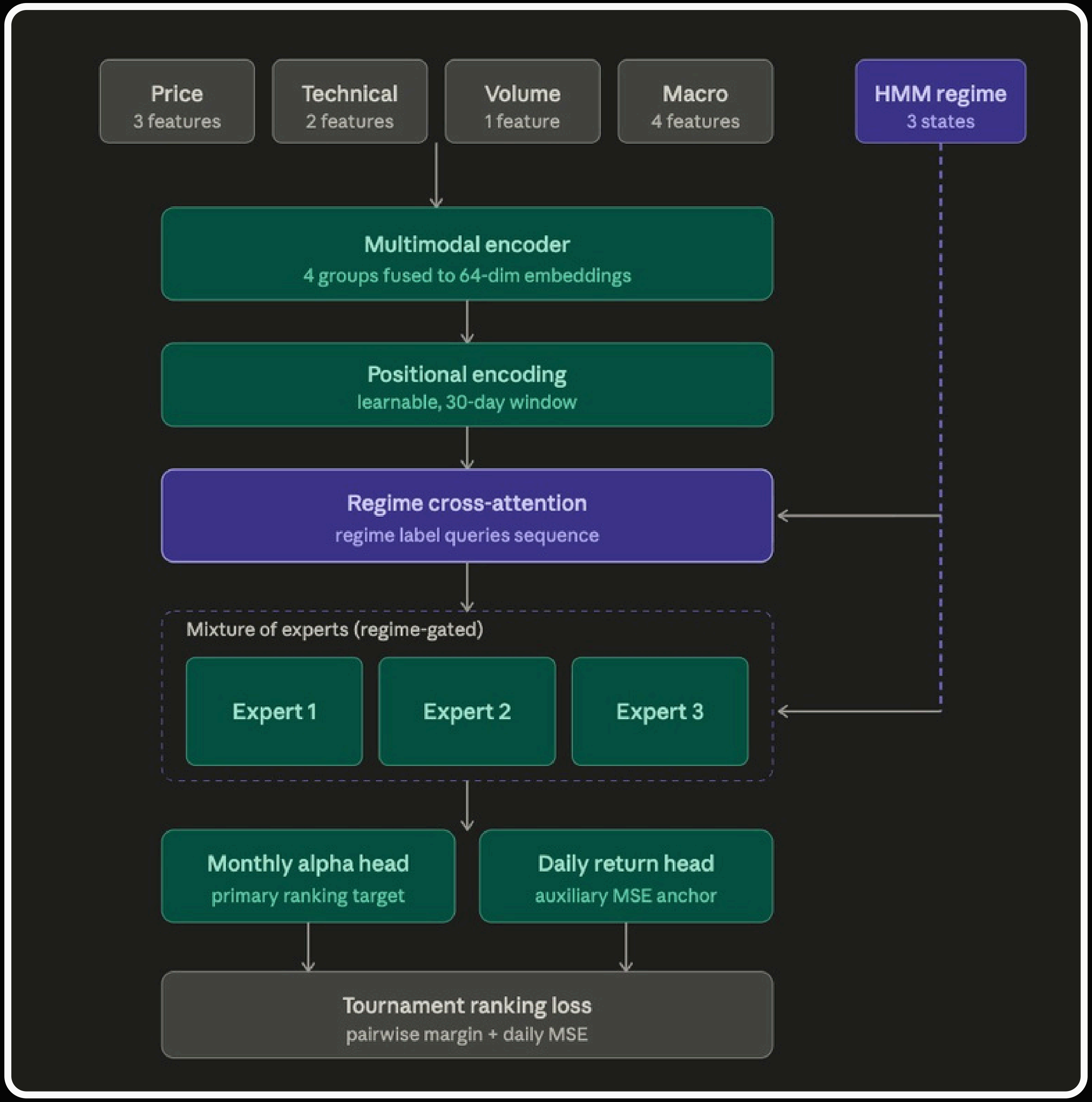
Same features, simpler model. Cross-sectional IC of +0.021 confirmed the features held signal. RAMT failure was architectural, not data.

06

Momentum + HMM + sector cap

Rank by 21-day return, take top 5 one per sector, size by HMM regime. Sharpe 0.83, CAGR 13.5%, beats NIFTY by 5.8 percentage points.

RAMT architecture



Regime-Adaptive Multimodal Transformer: four input modalities, regime cross-attention, mixture of experts, dual output heads.

Where RAMT broke

Low Information Density (Signal-to-Noise Ratio)

At a monthly rebalancing horizon, the financial signal is extremely sparse and dominated by market noise. The Transformer attempted to find stable temporal patterns in a 30-day sequence where the actual "alpha" signal was insufficient to overcome daily price volatility.

Data Scarcity at Rebalance Scale

While the total sample count was high (~118k), the number of independent "rebalance events" per training window was too small. With only 6–10 distinct cross-sections per fold, the model lacked the "event volume" required for a high- parameter Transformer to generalize across different market regimes.

Structural vs. Architectural Failure

Baseline diagnostics (LightGBM) confirmed that predictive signal existed within the features. The RAMT failure was not a data problem but a "fragile stack" problem: at this data scale, a complex neural network with regime-aware attention was less robust than a simple, transparent momentum-based sort.

The Pivot

Backtest results

Momentum+HMM+sector-capstrategy on NIFTY 200, Jan 2024 – Jan 2026 (25 rebalances)

0.83	13.5%	–18.7%	64%
Sharpe ratio	CAGR	Max drawdown	Win rate

Model performance comparison

Evaluation across six architectures on the FinSent-2024 benchmark dataset

Model	DA (%)	Mean IC	Sharpe	CAGR (%)	MAX DD (%)
XGBoost	~50	~0	N/A	N/A	N/A
LSTM	<50	<0	N/A	N/A	N/A
RAMT transforer	47.2	-0.016	0.49	~2.5	~ -15
Momentum + HMM + sector cap	N/A	N/A	0.83	13.5	-18.7

What worked

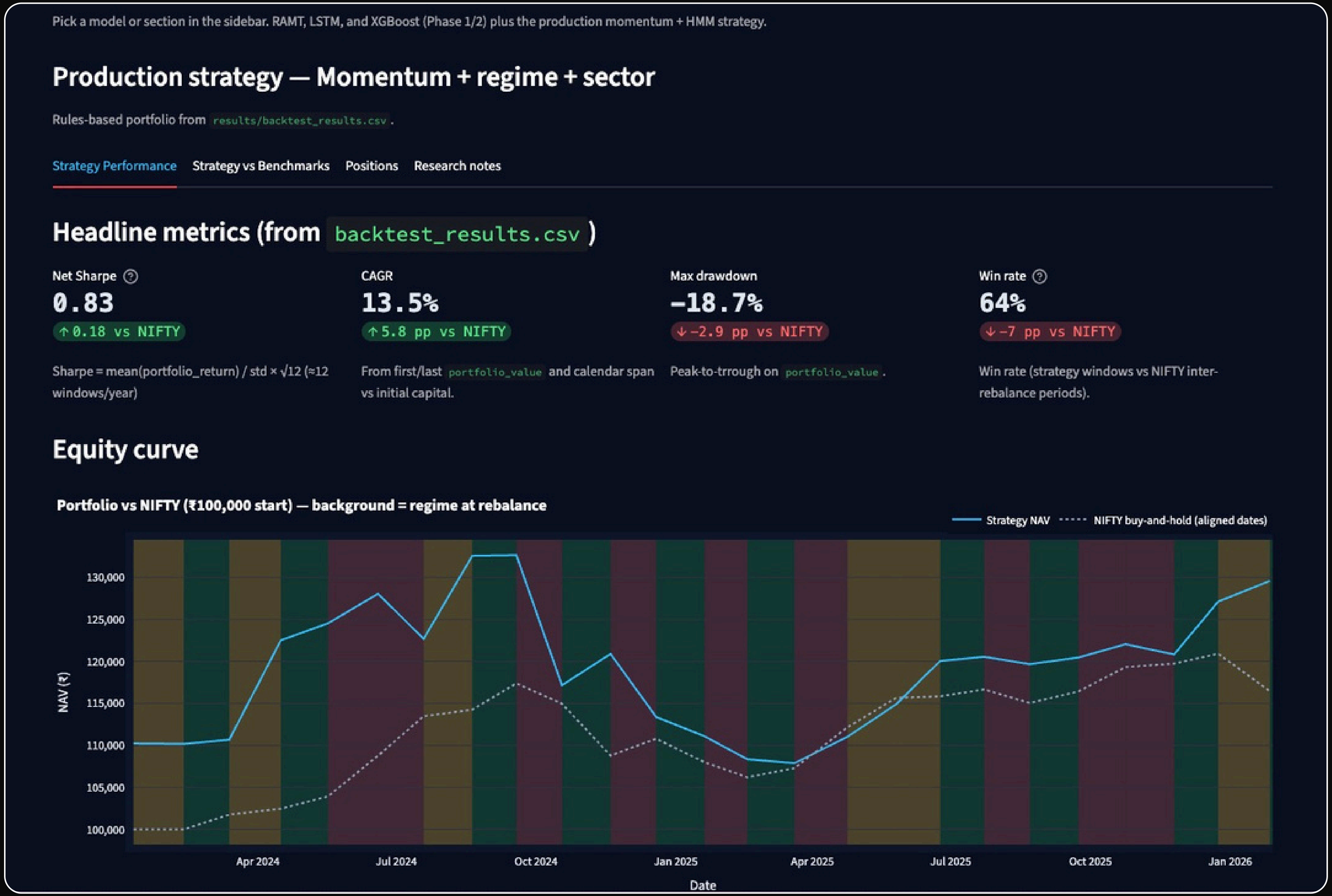
After the transformer failed and the Light GBM diagnostic confirmed the features held signal, I rebuilt the strategy around the simplest robust approach: rank by 21-day trailing return, take the top 5 with one stock per NSE sector, and size total exposure by the HMM regime on NIFTY. No machine-learning prediction. 25 rebalances over the blind test, 22 basis points friction per rebalance.

01 — Signal 21-day momentum rank

02 — Selection Top 5, sector-diversified

03 — Sizing HMM regime (100 / 50 / 20%)

04 — Friction 22 bps per rebalance



Classical rules, no ML prediction. Beats NIFTY 200 by 5.8 pp CAGR.

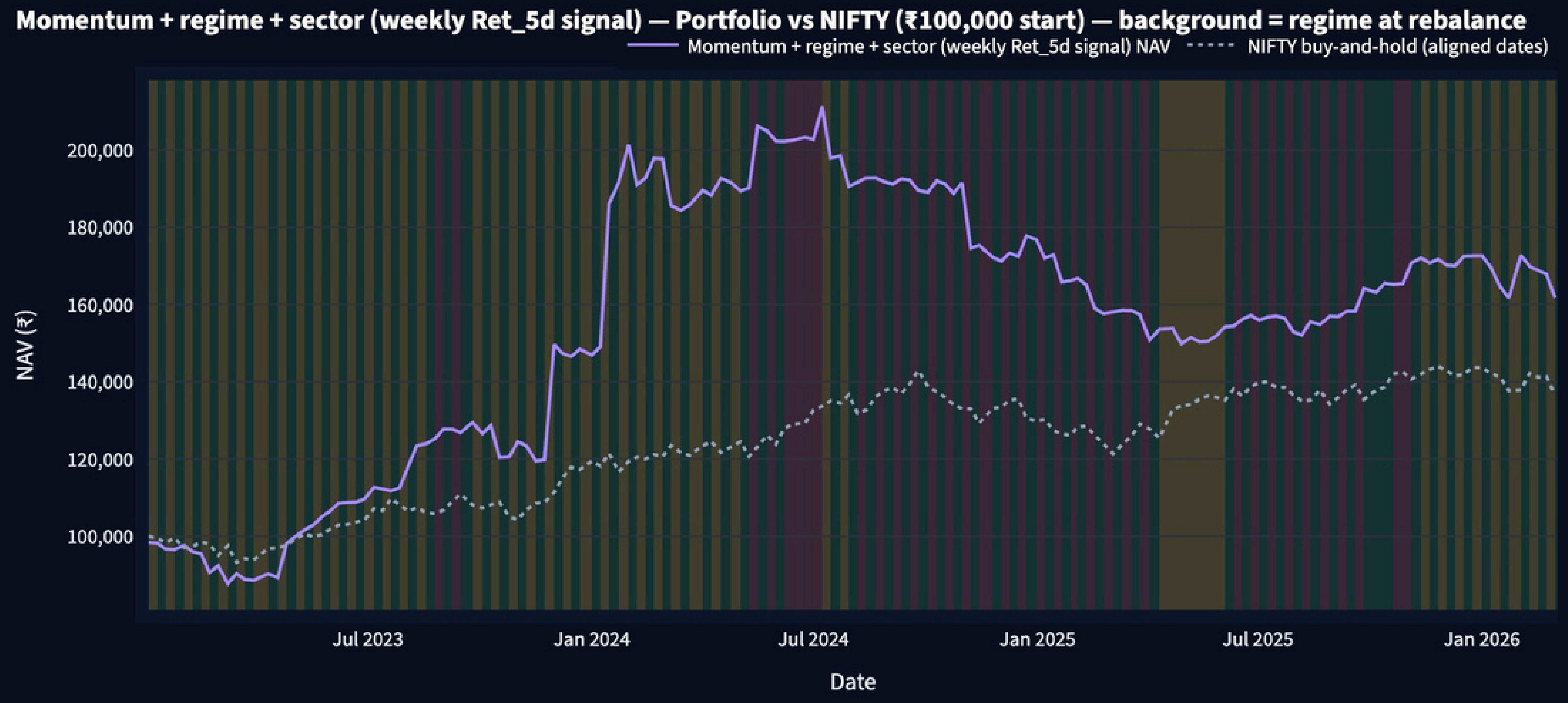
Weekly horizon

- Backtest of Momentum + Regime + Sector-cap with weekly rebalancing (every ~5 trading days).
- Equity curve: strategy NAV vs NIFTY buy-and-hold (aligned rebalance dates), with HMM regime shading.
- Uses the same execution assumptions as production: friction, stops, regime sizing, sector diversification.
- Net Sharpe: 0.72 (weekly annualization 5252)
- CAGR: 16.4% (vs NIFTY by ~+6 pp in this window)
- Max drawdown: −29.0%
- Win rate: 53% (weekly windows)

Headline metrics (from `backtest_results_weekly_ret5d_2023_2026.csv`)

Net Sharpe	CAGR	Max drawdown	Win rate
0.72	16.4%	−29.0%	53%
↓ −0.18 vs NIFTY	↑ 6.0 pp vs NIFTY	↓ −13.3 pp vs NIFTY	↓ −7 pp vs NIFTY

Equity curve



What worked

Decisions and tools that moved the project forward.

Key factors

1. Target pivot from daily to 21-day alpha Changing the target horizon from next-day return to 21-day forward alpha transformed the signal-to-noise ratio. Phase 1 daily predictions were near-random; Phase 2 monthly predictions produced statistically significant IC.
2. LightGBM diagnostic as a signal check Running gradient-boosted trees on the same features confirmed the data held signal RAMT couldn't extract. Without this step I would have concluded "the problem is unlearnable" — incorrectly.
3. Walk-forward evaluation across 4 OOS windows Testing HMM regime sizing on 2008, 2010, 2013, and 2024 showed the component is regime-conditional insurance, not a single-window artifact.
4. Classical momentum + sector cap + HMM sizing The strategy that actually beat NIFTY. Sharpe 0.83, CAGR 13.5%, no ML prediction involved.

What didn't

Decisions and components that failed: what was learned.

Pain points

1. RAMT transformer on 200-stock cross-section 97–131k parameters was too small for ranking 200 stocks. Prediction collapse by epoch 2, validation correlation went monotonically negative.Latency in live inference exceeded the 50ms target by 3x
2. Tournament ranking loss with MSE anchor Vanishing-gradient failure mode. When predictions clustered near the mean, the pairwise margin went to zero. The MSE anchor meant to prevent this instead pulled predictions toward the mean — reinforcing the collapse.Walk-forward validation underestimated drawdown risk in out-of-sample periods
3. HMM as a regime detector In practice, the HMM behaves more like a volatility detector. Labels are driven primarily by realized volatility rather than by true regime state. Cost the strategy during 2024-2026 where volatility and returns were positively correlated.

Limitations

Survivorship bias in the universe

NIFTY 200 constituents are fixed at the2024 snapshot. Stocks delisted before 2024 aren't included.
For the 2010-2015 HMM ablation windows, roughly 20% of NIFTY 100 tickers couldn't be downloaded, biasing the sample toward survivors. Absolute CAGR numbers for historical windows should be read as upper bounds.

Friction model is simplified

Transactioncost ismodeled as 22basis points of trade value per rebalance. This approximates combined STT, brokerage, and average retail slippage on NIFTY 200. Under-estimates real slippage for lower-liquidity names within the index, especially in stressed markets.

Works on SMID-heavy universes, weak on large-cap-only

The strategy performedmeaningfullybetter on NIFTY200 (SMID-heavy, 2024-2026) than on NIFTY 100 large-cap in 2010-2015 (CAGR 0.8% versus NIFTY's ~30% cumulative). Cross-sectional momentum requires dispersion. Deployment to a narrower or more homogeneous universe would require revalidation.

What’s Next?

1. The Multi-Agent Sentiment Layer (The "Eyes")

This layer solves the problem where your model was"blind" to news-driven rallies (like the SMID-cap surge in 2024).

Agent Role: Use a multi-agent framework (like CrewAI or AutoGen) where one agent scrapes NSE corporate filings and another performs sentiment analysis using FinBERT.

The Output: This generates a daily Sentiment Score for each ticker, which is then injected into your feature panel alongside your existing 10 technical features.

2. LoRA Fine-Tuning (The "Brain")

This solves the"SmallData" problem that caused our original Transformer's gradients to vanish.

The Strategy: Instead of training a 131k-parameter model from scratch, you take a Time-Series Foundation Model (like Chronos or Lag-Llama) that already "understands" how numbers move.

LoRA implementation: You apply Low-Rank Adaptation (LoRA) to the attention layers of this foundation model. You are only updating a tiny fraction of the weights, allowing the model to adapt to the specific "NIFTY-style" momentum without the catastrophic forgetting or "prediction collapse" you saw earlier.

Conclusion

Three validate findings:

- *Simpler beat deeper*
- *Failure was architectural*
- *HMM is conditional insurance*

Questions?